

Statistical Methods in Chemistry

Block 6

Reporting Chemical Results

Student handout

Name: _____

Class: _____

Date: _____

Contents

1	What a chemical result must communicate	2
2	Rounding, units, and uncertainty notation	2
3	Tables: raw data and processed data	3
4	Graphs: from data display to chemical argument	4
5	Error bars and visual claims	5
6	Reporting data rejection	6
7	Writing conclusions from data	7
8	Limitations and improvements	8
9	Mini-practical report task	8
10	Final reporting checklist	10
	Answer key	11

1 What a chemical result must communicate

A chemical result is not just a number. A useful result tells the reader what was measured, how it was measured, how uncertain the result is, and what conclusion can be drawn from it.

Core idea

A reported chemical result should be complete enough that another chemist can judge its reliability without repeating the whole experiment.

A defensible result usually includes:

Component	Example
Measured quantity	Concentration of CH_3COOH in diluted vinegar
Value and unit	$0.737 \text{ mol dm}^{-3}$
Uncertainty spread	or $(0.737 \pm 0.006) \text{ mol dm}^{-3}$, or $s = 0.004 \text{ mol dm}^{-3}$
Sample size	$n = 4$ concordant titres
Method	Acid-base titration using standardized NaOH
Conditions	Temperature, wavelength, indicator, dilution, pH, or other relevant conditions
Conclusion	Whether the value supports the chemical claim or agrees with a reference value

A weak report says: “The concentration was 0.737.” A stronger report says: “The concentration of ethanoic acid in the original vinegar was reported as $(0.737 \pm 0.006) \text{ mol dm}^{-3}$ from four concordant titres after a ten-fold dilution.”

Quick exercise: Complete or incomplete?

For each result, state what information is missing.

a) “The concentration was $0.104 \text{ mol dm}^{-3}$.”

b) “The mean titre was 24.62 cm^3 , so the method worked.”

c) “The sample contained 4.8 g dm^{-3} copper.”

2 Rounding, units, and uncertainty notation

Chemical reports should not show more precision than the experiment supports. Excess decimal places make a result look more accurate than it is.

Reporting with uncertainty

A common school-level rule is:

- 1) Round the uncertainty to one significant figure, except when a teacher or specification asks otherwise.
- 2) Round the measured value to the same decimal place as the uncertainty.
- 3) Always include units.

Worked example: Rounding a concentration

A calculation gives:

$$c = 0.104\,23 \text{ mol dm}^{-3}, \quad u = 0.0031 \text{ mol dm}^{-3}.$$

Round the uncertainty first:

$$0.0031 \text{ mol dm}^{-3} \approx 0.003 \text{ mol dm}^{-3}.$$

Then round the concentration to the same decimal place:

$$c = (0.104 \pm 0.003) \text{ mol dm}^{-3}.$$

Poor reporting	Problem	Better reporting
0.104231	No unit; too many digits	$0.104 \text{ mol dm}^{-3}$
24.6 ml	Wrong unit style; unclear precision	24.60 cm^3
5 grams/litre	Informal unit expression	5.0 g dm^{-3}
0.102 ± 0.0000003	Unrealistic uncertainty for a school titration	Check uncertainty calculation

Quick exercise: Round the results correctly

Rewrite each result in a defensible form.

a) $24.681 \text{ cm}^3 \pm 0.064 \text{ cm}^3$

b) $0.098\,734 \text{ mol dm}^{-3} \pm 0.0028 \text{ mol dm}^{-3}$

c) $1.237 \cdot 10^{-3} \text{ mol} \pm 4.9 \cdot 10^{-5} \text{ mol}$

3 Tables: raw data and processed data

A good table lets the reader check the logic of the analysis. Raw data and processed data should usually be separated. Raw data are direct measurements. Processed data are values calculated from measurements.

Rules for chemical tables

- Put the quantity and unit in the column heading, not in every cell.
- Use consistent decimal places for repeated measurements made using the same instrument.
- Do not mix raw readings and calculated values without clear labels.
- Include uncertainty information when it is relevant.
- Use enough digits for checking, but not meaningless extra digits.

Trial	Initial burette / cm^3	Final burette / cm^3	Titre / cm^3	Use in mean?
Rough	0.10	18.90	18.80	No
1	0.20	18.62	18.42	Yes
2	0.05	18.51	18.46	Yes
3	0.15	18.58	18.43	Yes
4	0.00	18.45	18.45	Yes

Worked example: Processed-data table

Processed quantity	Value
Mean titre used	18.44 cm ³
Sample standard deviation	0.018 cm ³
Number of titres in mean	$n = 4$
Relative standard deviation	0.10%

Quick exercise: Find the table problems

A student reports the following table.

Trial	volume	concentration	comment
1	24.6	0.100341231	good
2	24.61 ml	0.100382144	good
3	24.7	0.100749001	close enough

List three problems with this table.

4 Graphs: from data display to chemical argument

A graph in a chemistry report is not decoration. It should show the relationship being tested and help the reader judge whether the conclusion follows from the data.

Minimum requirements for a chemistry graph

A useful graph should include labelled axes with units, appropriate data markers, a justified line or curve, uncertainty information when relevant, and a caption that explains the chemical meaning of the figure.

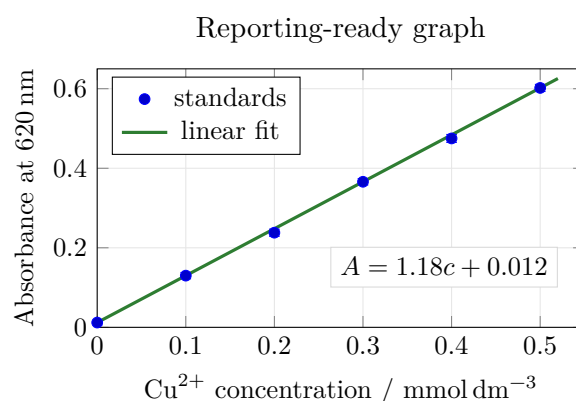
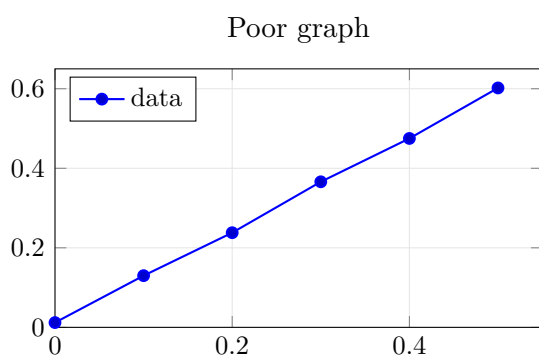


Figure 1: The same calibration data can be displayed poorly or in a form that supports chemical interpretation.

The poor graph has no axis labels or units and connects points as if the order of measurement were chemically meaningful. The improved graph states the measured quantities, shows error bars, and uses a line only because a linear calibration model is chemically justified.

Quick exercise: Graph checklist

A student creates a calibration graph but forgets to include the wavelength, units, error bars, and equation of the line. Explain why each missing item matters.

5 Error bars and visual claims

Error bars are not decorative. They show the uncertainty or spread associated with a plotted value. A conclusion should not ignore them.

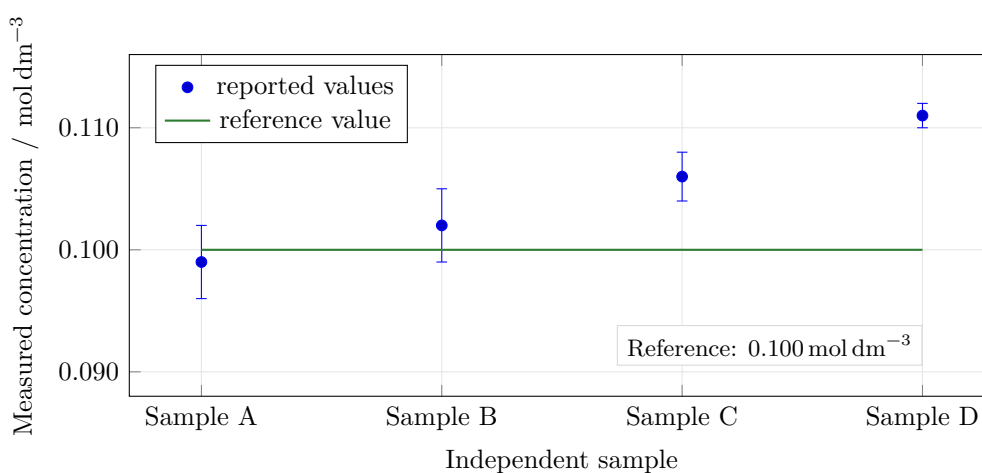


Figure 2: A report should interpret values together with their uncertainty intervals.

In the figure, Sample A agrees well with the reference value. Sample B is slightly high but its uncertainty interval still includes the reference. Sample C is probably high. Sample D is clearly high and precise, which suggests a systematic problem or a genuinely different sample.

Do not overclaim from a graph

A graph can suggest a pattern, but the written conclusion must explain what the graph supports and what it does not prove. A small visual difference may not be meaningful if uncertainty intervals are large.

Quick exercise: Interpret error bars

Using the figure above, answer the questions.

a) Which sample agrees best with the reference value?

b) Which sample is most clearly above the reference value?

c) Which sample has a high central value but still may be compatible with the reference value?

6 Reporting data rejection

Data rejection must be justified. A report should not simply remove inconvenient data because they make the result look worse.

Rule for rejecting data

A data point may be excluded only when there is a defensible reason, such as an identified procedural mistake, an instrument problem, or a pre-defined statistical rule. The reason should be stated in the report.

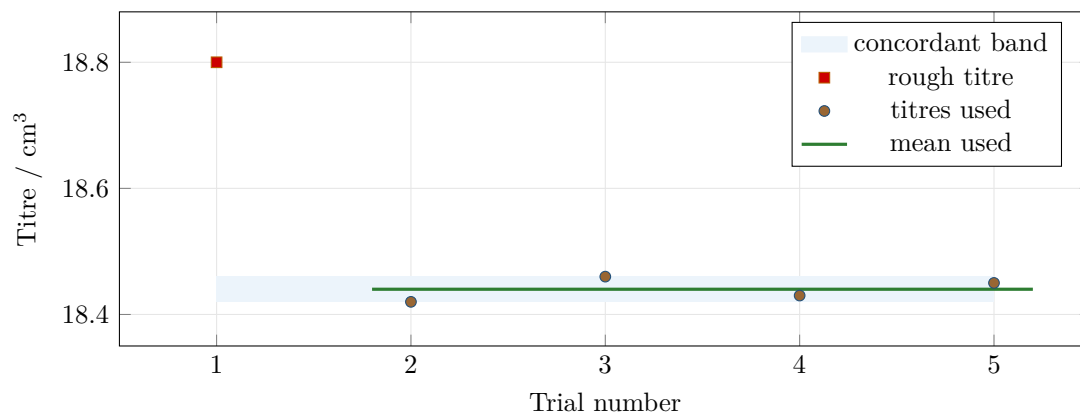


Figure 3: A rough titre can be excluded if the report clearly states that it was exploratory and not used in the final mean.

A correct report might say: “The rough titre was used to locate the endpoint and was not included in the mean. Trials 1–4 after the rough titre were concordant within 0.04 cm^3 ; therefore their mean, 18.44 cm^3 , was used for the final calculation.”

A weak report says: “The first result was ignored because it was different.” That is not enough.

Quick exercise: Reject or keep?

For each case, decide whether exclusion is justified. Give a reason.

- a) A first titration was performed quickly to estimate the endpoint and gave 25.80 cm^3 ; the next three titres were 24.62 cm^3 , 24.64 cm^3 , and 24.63 cm^3 .

- b) One absorbance point lies below the calibration line, but the student has no record of a mistake.

- c) A pH probe reading was taken before calibration and then repeated after calibration.

7 Writing conclusions from data

The conclusion is where statistical evidence is converted into chemical meaning. It should not merely repeat calculations.

Claim - evidence - reasoning

A strong conclusion has three parts:

- 1) **Claim:** the answer to the research question.
- 2) **Evidence:** numerical result, uncertainty, graph, test, or comparison.
- 3) **Reasoning:** why the evidence supports the claim chemically.

Conclusion template

Claim: The sample concentration was estimated to be _____.

Evidence: This value was obtained from _____ trials / a calibration curve / a validated method, with _____ uncertainty.

Comparison: Compared with the reference value of _____, the result is _____.

Reasoning: The result is reliable / limited because _____.

Chemical meaning: This suggests that _____.

Worked example: From weak to strong conclusion

Weak: "The experiment was successful because the values were close."

Stronger: "The concentration of the prepared NaOH solution was $(0.098 \pm 0.003) \text{ mol dm}^{-3}$, based on four concordant titrations. The accepted value was $0.100 \text{ mol dm}^{-3}$, which lies inside the uncertainty interval. Therefore, the data are consistent with the expected concentration, although the method cannot distinguish small differences below approximately $0.003 \text{ mol dm}^{-3}$."

Quick exercise: Improve the conclusion

Rewrite this conclusion so that it uses claim, evidence, and reasoning.

"The copper solution had a concentration of 0.35, and the graph was quite straight, so the experiment was good."

8 Limitations and improvements

A limitation is not the same as a personal excuse. Reports should avoid vague phrases such as “human error” or “we were not careful enough.” A useful limitation identifies a specific source of uncertainty or bias.

Weak limitation	Better limitation	Better improvement
Human error	Endpoint colour change was subjective near pale pink	Use a pH probe or a more suitable indicator
Not enough time	Only two repeats were obtained, so precision is uncertain	Repeat until at least three concordant titres are obtained
Equipment was bad	Balance resolution was 0.01 g, large relative to sample mass	Use a larger sample mass or a 0.001 g balance
Graph was not perfect	High concentration standards deviated from linearity	Dilute samples and restrict calibration to the linear range

Improvement rule

An improvement should target a named limitation. If the limitation is random scatter, add repeats or improve precision. If the limitation is systematic bias, change calibration, standards, blank correction, or method design.

Quick exercise: Limitation or excuse?

For each statement, rewrite it as a useful limitation and propose a specific improvement.

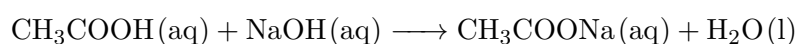
- a) “The results were bad because the group made mistakes.”

- b) “The graph was not very good.”

- c) “The balance was not accurate.”

9 Mini-practical report task

A student determined the concentration of ethanoic acid in vinegar by titration. A 10.00 cm³ sample of vinegar was diluted to 100.0 cm³. Then 25.00 cm³ of the diluted vinegar was titrated with 0.1000 mol dm^{−3} sodium hydroxide.



Trial	Initial burette / cm ³	Final burette / cm ³	Titre / cm ³	Comment
Rough	0.10	18.90	18.80	Endpoint search
1	0.20	18.62	18.42	Use
2	0.05	18.51	18.46	Use
3	0.15	18.58	18.43	Use
4	0.00	18.45	18.45	Use

Quick exercise: Prepare the report section

Complete the reporting tasks.

- 1) Calculate the mean titre used.

- 2) Calculate the concentration of ethanoic acid in the diluted vinegar.

- 3) Calculate the concentration in the original vinegar.

- 4) Report the final concentration to a sensible number of significant figures.

- 5) Write a two-sentence conclusion using claim, evidence, and reasoning.

- 6) State one limitation and one targeted improvement.

10 Final reporting checklist

Before submitting a chemistry report, check that every result is interpretable.

Check	Question
☐	Does every numerical result have a unit?
☐	Are uncertainties, standard deviations, or confidence intervals reported where relevant?
☐	Are values rounded to a precision supported by the measurements?
☐	Are raw data and processed data clearly separated?
☐	Do graphs have labelled axes, units, captions, and justified trend lines?
☐	Are any excluded data points justified before they are removed?
☐	Does the conclusion use numerical evidence rather than vague claims?
☐	Are limitations specific and linked to targeted improvements?

One-sentence standard

A good chemistry report should allow the reader to reconstruct the logic from raw measurements to chemical conclusion.

Answer key

Complete or incomplete?

- a) Missing uncertainty/spread, number of repeats, and method context.
- b) Missing standard deviation or uncertainty, acceptance criterion, and explanation of why the method worked.
- c) Missing method, sample identity or matrix, uncertainty, and possibly whether this was copper ions, copper metal equivalent, or a copper compound.

Round the results correctly

- a) $(24.68 \pm 0.06) \text{ cm}^3$.
- b) $(0.099 \pm 0.003) \text{ mol dm}^{-3}$.
- c) $(1.24 \pm 0.05) \times 10^{-3} \text{ mol}$, or $(0.00124 \pm 0.00005) \text{ mol}$.

Find the table problems

Possible answers: units are missing from the headings; units are mixed into one cell; volume decimal places are inconsistent; concentration has meaningless extra digits; “close enough” is not a criterion; it is unclear whether concentration is raw or processed data.

Graph checklist

The wavelength matters because absorbance depends on wavelength. Units are needed so that concentration can be interpreted. Error bars show measurement spread or uncertainty. The equation is needed if the graph is used to calculate an unknown concentration.

Interpret error bars

- a) Sample A agrees best with the reference.
- b) Sample D is most clearly above the reference.
- c) Sample B has a high central value but may still be compatible with the reference because its uncertainty interval includes $0.100 \text{ mol dm}^{-3}$.

Reject or keep?

- a) Exclusion is justified if it was explicitly a rough titre used only to locate the endpoint.
- b) Do not exclude without a recorded mistake or pre-defined rule.
- c) Excluding the pre-calibration value is justified because the instrument was not yet calibrated.

Improve the conclusion

Example: “The concentration of the copper solution was estimated as $0.35 \text{ mmol dm}^{-3}$ from the calibration curve. The result is supported by the near-linear calibration trend, but the conclusion should also report the equation, uncertainty, and whether the unknown lies inside the calibrated range.”

Limitation or excuse?

Examples:

- a) Limitation: the endpoint colour change was difficult to judge consistently. Improvement: use a pH probe or a more suitable indicator.
- b) Limitation: the calibration graph shows curvature above a certain concentration. Improvement: dilute standards and samples into the linear range.
- c) Limitation: the balance resolution was too low relative to the sample mass. Improvement: use a larger sample mass or a more precise balance.

Mini-practical report task

Mean titre:

$$\bar{V} = \frac{18.42 + 18.46 + 18.43 + 18.45}{4} = 18.44 \text{ cm}^3.$$

Moles of NaOH used:

$$n(\text{NaOH}) = 0.1000 \times \frac{18.44}{1000} = 1.844 \cdot 10^{-3} \text{ mol}.$$

The mole ratio is 1:1, so $n(\text{CH}_3\text{COOH}) = 1.844 \cdot 10^{-3} \text{ mol}$ in 25.00 cm^3 of diluted vinegar.

$$c_{\text{diluted}} = \frac{1.844 \times 10^{-3}}{0.02500} = 0.07376 \text{ mol dm}^{-3}.$$

The original vinegar was diluted ten-fold:

$$c_{\text{original}} = 10 \times 0.07376 = 0.7376 \text{ mol dm}^{-3} \approx 0.738 \text{ mol dm}^{-3}.$$

A suitable conclusion: “The concentration of ethanoic acid in the original vinegar was $0.738 \text{ mol dm}^{-3}$, based on four concordant titres with a mean titre of 18.44 cm^3 . The rough titre was excluded because it was used only to locate the endpoint; the final result is supported by the close agreement among the four analytical titres.”